Multi-Joint Reaching,
Muscle Synergies & PCA

From Single-Joint to Center-Out Reaching — Dimensionality Reduction in Motor Control

Week 4 Lecture — Machine Learning for Neuroscience

Unsupervised Learning for Multi-Muscle Analysis

Lecture Outline

1. From Single Joint to Multi-Joint Reaching

In Weeks 1-3, we modeled a single elbow joint: one degree of freedom, driven by agonist and antagonist muscles. Real reaching involves coordinating **multiple joints** simultaneously. The simplest multi-joint model of the arm is a **2-DOF shoulder-elbow planar linkage**, where the shoulder angle q_1 and elbow angle q_2 together determine the hand position in Cartesian space via **forward kinematics**.

Forward kinematics:

$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$
$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

where L_1 = 0.30 m (upper arm) and L_2 = 0.35 m (forearm). The critical new feature of multi-joint movement is **interaction torques**: motion at one joint creates Coriolis and centripetal forces that perturb the other joint. The nervous system must predict and compensate for these mechanical interactions, which is one reason multi-joint coordination is computationally harder than single-joint control.

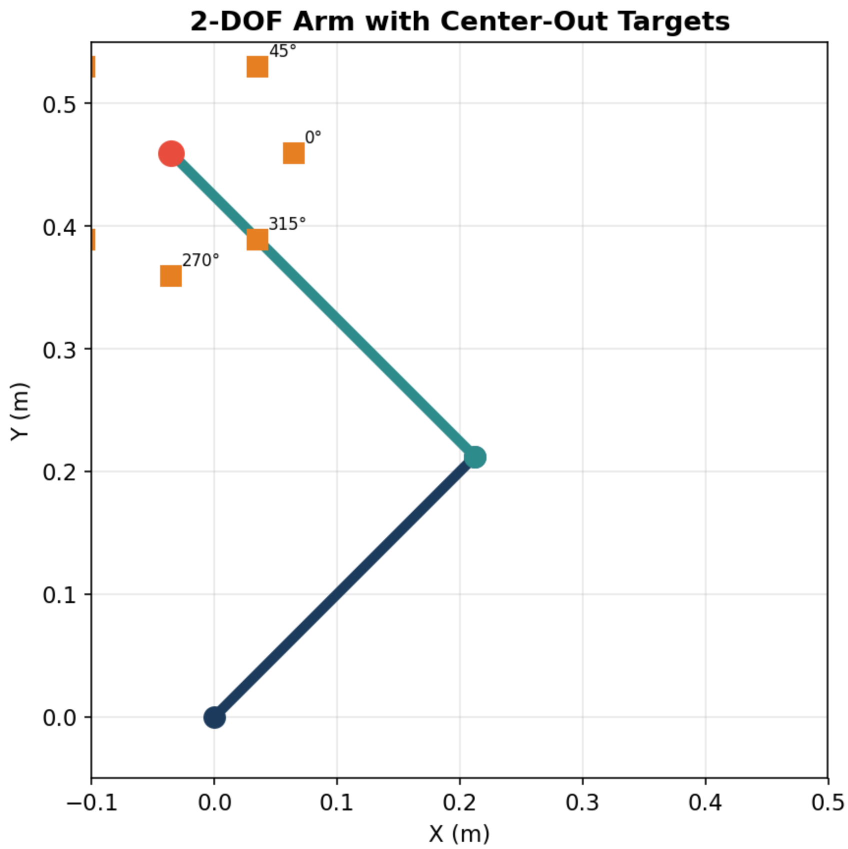

Figure 1. 2-DOF shoulder-elbow arm with 8 center-out targets at 10 cm distance. The arm is drawn at its rest configuration (q1 = 60°, q2 = 70°), with target squares distributed radially from the hand position.

Key Concept: Forward kinematics maps joint angles to hand position. The inverse mapping (hand position to joint angles) is not unique — multiple joint configurations can reach the same point (the *inverse kinematics problem*).

2. Bernstein's Degrees-of-Freedom Problem

Nikolai Bernstein (1967) identified a fundamental puzzle: the human motor system has far more **degrees of freedom** (muscles, joints, motor units) than needed for any given task. A 2-DOF arm is controlled by at least 6 muscles (shoulder flexor/extensor, elbow flexor/extensor, and biarticular flexor/extensor). This means there are **infinitely many** possible muscle activation patterns that could produce the same joint torques — the system is **redundant**.

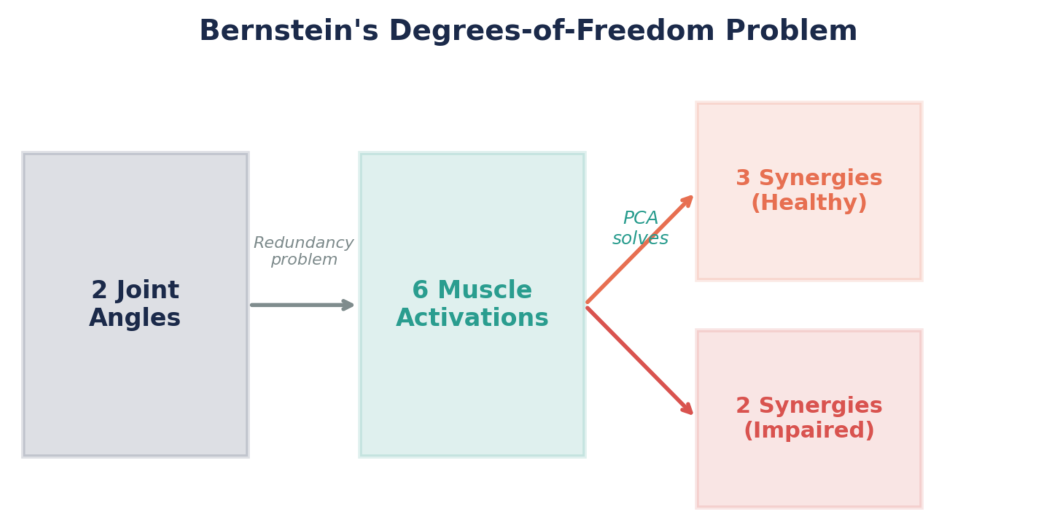

Figure 2. Bernstein's degrees-of-freedom problem: 2 joint DOFs are controlled by 6 muscles, creating infinite possible activation patterns. Muscle synergies reduce this to 2-3 dimensions.

How does the CNS choose among the infinitely many solutions? One influential hypothesis is that the brain controls not individual muscles, but **muscle synergies** — coordinated patterns of muscle co-activation that act as low-dimensional building blocks. If 6 muscles are controlled via 2 synergies, the control problem reduces from 6 dimensions to 2, dramatically simplifying the computational burden.

Bernstein's Insight: Motor redundancy is both a curse (too many choices) and a blessing (flexibility to adapt). Synergies are one way the CNS tames redundancy by reducing the effective dimensionality of motor control.

3. Center-Out Reaching Task

The center-out reaching paradigm (Georgopoulos et al., 1982) is the workhorse experiment for studying directional motor control. The subject starts with their hand at a central position and reaches to one of several targets arranged in a circle. With **8 equally spaced targets** (0°, 45°, 90°, ... 315°), the task samples the full range of reaching directions.

Our dataset:

- **Subjects:** 20 total (10 healthy controls, 10 impaired)
- **Targets:** 8 directions (every 45°)
- **Speeds:** 3 conditions (slow: 700 ms, medium: 500 ms, fast: 300 ms)
- **Muscles:** 6 recorded per trial (EMG)
- **Total:** 20 × 8 × 3 = 480 trials, each with 6-muscle EMG time series

Each muscle has a **preferred direction (PD)** — the reaching direction that elicits maximal activation. Muscle activation follows a **cosine tuning** model: activation is highest when the reach direction aligns with the muscle's PD and decreases as a cosine function for off-axis directions. This is analogous to the cosine tuning of motor cortical neurons described by Georgopoulos.

Dataset Structure: For each trial, the raw EMG is full-wave rectified and low-pass filtered to obtain the linear envelope. The peak envelope amplitude for each muscle is then extracted as a single scalar, yielding a 6-dimensional feature vector per trial. PCA is applied to this (trials × 6) matrix — not to the time series directly.

4. Multi-Muscle Activation Patterns

Our 6 muscles are arranged in three agonist-antagonist pairs, acting in the **horizontal plane** (the arm moves on a frictionless table surface). Shoulder muscles produce horizontal flexion (across the body) and extension (away from the body):

Muscle	Abbreviation	Function	Crosses
Shoulder H. Flexor	Sh.H.Flex	Moves arm across body (leftward)	Shoulder only
Shoulder H. Extensor	Sh.H.Ext	Moves arm away from body (rightward)	Shoulder only
Elbow Flexor	El.Flex	Bends elbow	Elbow only
Elbow Extensor	El.Ext	Straightens elbow	Elbow only
Biarticular Flexor	Bi.Flex	Flexes both joints	Both joints
Biarticular Extensor	Bi.Ext	Extends both joints	Both joints

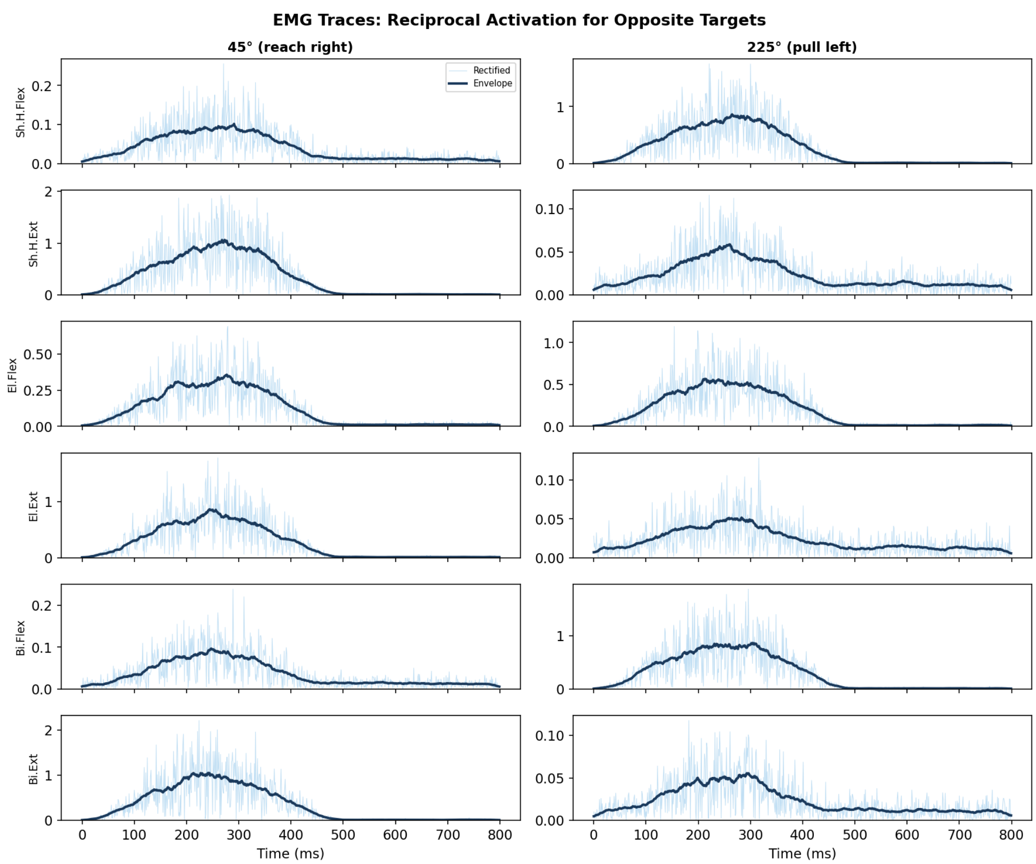

Figure 3. Full-wave rectified EMG (light traces) with low-pass filtered envelope (dark) for all 6 muscles during reaches to 45° (left column) and 225° (right column). Note the reciprocal activation pattern: muscles active for one direction are quiet for the opposite direction, confirming synergy-based generation.

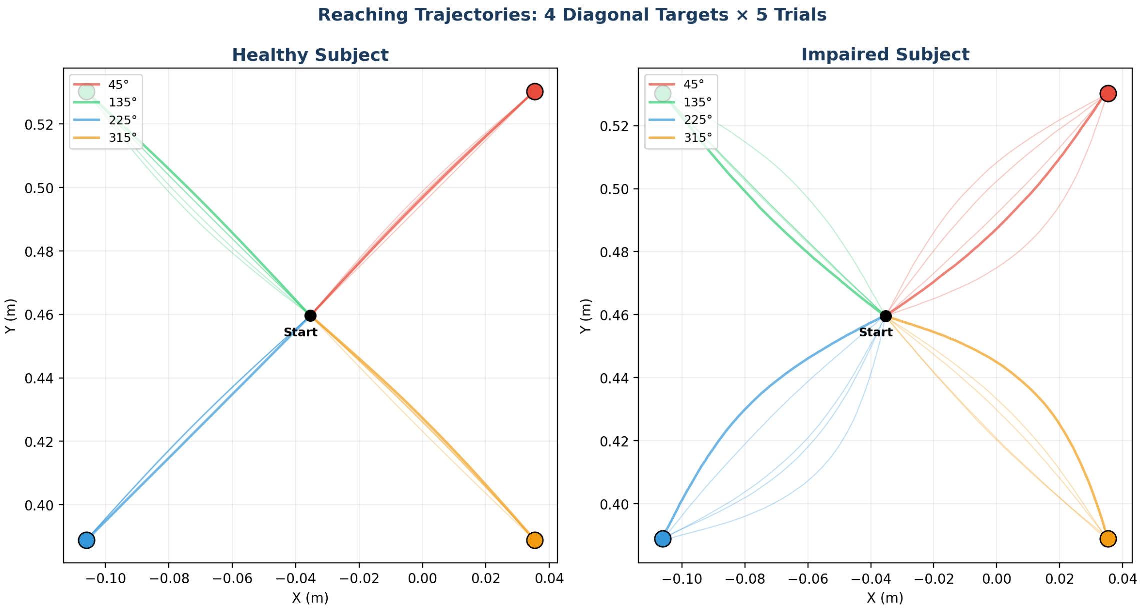

Figure 3b. Reaching trajectories for 4 diagonal targets (5 trials each). Left: healthy subject produces straight, tightly grouped hand paths. Right: impaired subject shows curved, more variable trajectories due to the merged synergy structure — fewer independent motor modules means less ability to make fine corrections mid-reach.

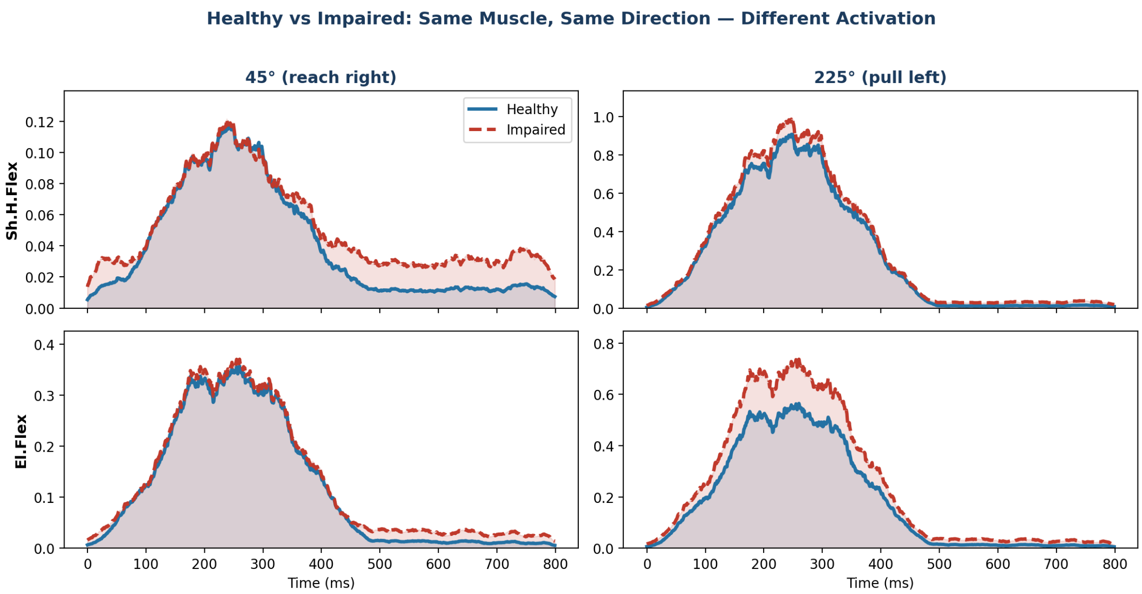

Figure 3c. EMG envelopes for two muscles (Sh.H.Flex, El.Flex) during reaches to 45° and 225°, overlaying healthy (blue, solid) and impaired (red, dashed). At 45°, the healthy shoulder flexor is nearly silent while the impaired version shows a large burst — this is co-contraction caused by synergy merging. The impaired nervous system cannot activate extensors without also recruiting flexors.

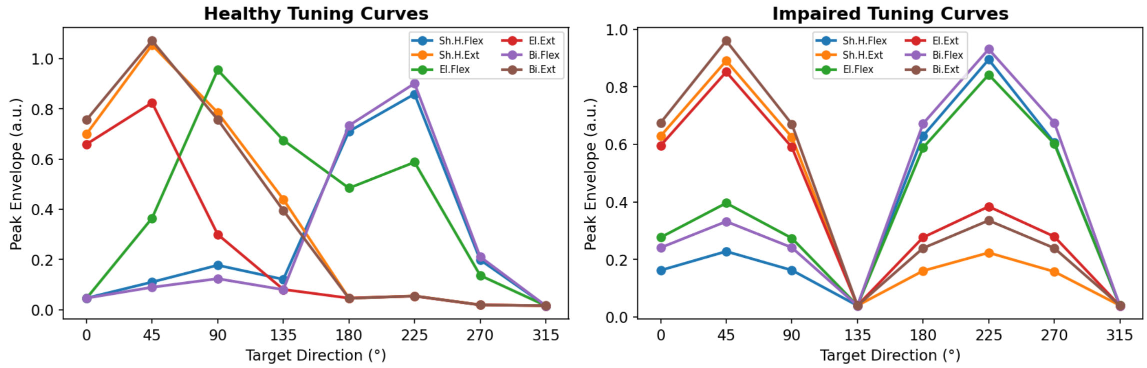

Figure 4. Directional tuning curves showing mean peak EMG envelope vs. target direction for all 6 muscles. Left: Healthy subjects show distinct preferred directions for each muscle. Right: Impaired subjects show similar directional preferences but with co-contraction (multiple muscles elevated simultaneously), reflecting fewer independent synergies.

In our synergy-based model, muscles are not controlled independently. Instead, **three underlying synergies** control the 6 muscles: (1) an **extensor synergy** co-activating Sh.H.Ext, El.Ext, Bi.Ext for reaching forward-right; (2) a **flexor synergy** co-activating Sh.H.Flex, El.Flex, Bi.Flex for pulling backward-left; and (3) an **elbow coordination synergy** coupling El.Flex with Sh.H.Ext for upward/diagonal reaches. This third synergy provides independent elbow control — and it is this module that is lost after neurological injury.

5. Principal Component Analysis: The Math

Principal Component Analysis (PCA) is an **unsupervised** dimensionality reduction technique that finds the directions of maximum variance in high-dimensional data. Given a data matrix **X** of shape (n_trials, n_muscles), PCA finds a new coordinate system where:

- The first axis (PC1) captures the **most variance** in the data
- PC2 captures the most remaining variance, orthogonal to PC1
- Each subsequent PC captures decreasing amounts of variance

The PCA algorithm:

Step 1. Center the data by subtracting the column means (and optionally scale to unit variance).

Step 2. Compute the covariance matrix: $C = (1/n)\ X^T X$, where C is (6×6) for 6 muscles. Alternatively, if the data have been standardized (zero mean, unit variance), the covariance matrix equals the **correlation matrix** R. Use the covariance matrix when variables share the same physical units and scale (e.g., all in millivolts); use the correlation matrix (equivalently, standardize first) when variables have different units or vastly different magnitudes, to prevent high-amplitude muscles from dominating the PCA. Since our 6 muscles can differ in raw amplitude, we standardize and thus work with the correlation matrix.

Step 3. Eigendecomposition: $C\, v_i = \lambda_i\, v_i$, where v_i is a single eigenvector (one principal component direction, length 6) and λ_i is its eigenvalue (the variance along that direction).

Step 4. Sort by eigenvalue ($\lambda_1 \geq \lambda_2 \geq \ldots$) and project: scores = $X\, V_k$, where V_k is the $(6 \times k)$ matrix whose **columns** are the top k eigenvectors $v_1, v_2, \ldots, v_k$. The lowercase v_i denotes a single eigenvector; the uppercase V_k collects k of them into a projection matrix.

Variance explained by PC i:

$$VE_i = \lambda_i / (\lambda_1 + \lambda_2 + \ldots + \lambda_p)$$

Improving interpretability with Varimax rotation. PCA maximizes variance but the resulting components can be hard to interpret because each PC may load on many variables. **Varimax rotation** rotates the PC axes (within the subspace spanned by the retained PCs) to produce a **simpler structure** where each variable loads strongly on fewer components. This does not change the total variance explained but redistributes it among the rotated components, often yielding loadings that more closely resemble the underlying synergies. We will not apply Varimax in this lab, but it is a common post-processing step in synergy research.

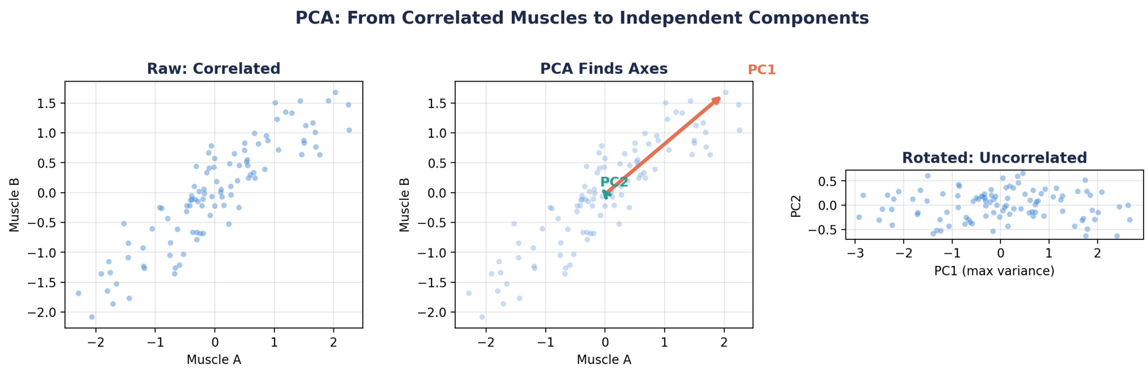

*Figure 5. PCA illustrated in 2D. Left: Raw data showing correlated muscle activations. Center: PCA finds the directions of maximum variance (PC1) and orthogonal complement (PC2). Right: Each data point is projected onto PC1 by computing its dot product with the PC1 eigenvector, yielding a single scalar score. **Interpreting signs:** A **positive score** means the trial's muscle pattern aligns with the PC direction (e.g., extensors ON, flexors OFF); a **negative score** means the opposite pattern (flexors ON, extensors OFF). For **loadings**: muscles with the same sign co-activate together within that synergy, while muscles with opposite signs are reciprocally activated. The magnitude indicates how strongly each muscle participates.*

Understanding the projection: To project a 2D data point onto PC1, we compute $score_i = x_i \cdot v_1$, the dot product of the data vector with the first eigenvector. Geometrically, this drops a perpendicular from the point onto the PC1 line. In the right panel, all points have been collapsed to this single axis. Information along the PC2 direction is discarded — but if PC1 captures most of the variance, little information is lost. Extending to 6 muscles: each trial's 6-muscle vector is projected onto a 2D plane spanned by v_1 and v_2, reducing 6 numbers to 2 synergy scores.

Key Insight: PCA rotates the coordinate system to align with the data's natural axes of variation. If most variance lies along 2-3 directions, then 6 muscles are effectively controlled by 2-3 synergies.

6. PCA Reveals Muscle Synergies

When we apply PCA to the mean EMG data from healthy subjects, a striking result emerges: **3 principal components are needed to explain over 90% of the variance** in healthy subjects. This means the 6-dimensional muscle activation space is effectively 3-dimensional — the CNS uses three independent motor modules, not six independent muscles.

EMG preprocessing for PCA: Since PCA operates on a (trials × muscles) matrix rather than on time series, we first reduce each muscle's EMG time series to a single scalar per trial. Here we use the **peak envelope amplitude** — the maximum of the low-pass filtered rectified EMG. Other common choices include integrated EMG (iEMG, the area under the envelope) or RMS amplitude. The choice affects the scale but not the synergy structure, because all three measures are monotonically related to the underlying muscle activation level.

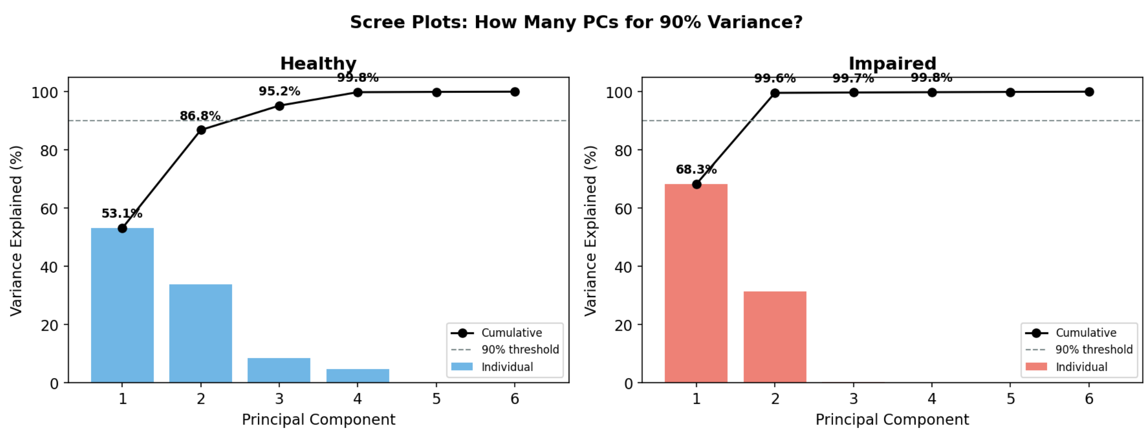

Figure 6. Scree plots for healthy and impaired subjects. Healthy: PC1 explains 53.1%, PC2 adds 33.7%, and PC3 adds 8.4%, reaching 95.2% cumulative (3 PCs needed for ≥90%). Impaired: PC1 alone explains 68.3% and PC2 adds 31.3%, reaching 99.6% with just 2 PCs. The impaired group's concentration of variance in fewer PCs reflects synergy merging.

Interpreting the loadings:

Each row of the **loadings matrix** (pca.components_) represents one synergy pattern. A **positive loading** means that muscle's activation increases when the synergy score is high; a **negative loading** means it decreases. Two muscles with the **same sign** (both positive or both negative) are **co-activated** by that synergy; muscles with **opposite signs** are **reciprocally activated** (one goes up while the other goes down). After **Varimax rotation** (which rotates PCs to maximize simple structure while preserving total variance), the loadings become easier to interpret as distinct muscle groupings.

In healthy subjects after Varimax rotation, three clear synergies emerge: **Synergy 1** loads on Sh.H.Ext, El.Ext and Bi.Ext (extensor group — reaching outward). **Synergy 2** loads on Sh.H.Flex, Bi.Flex and El.Flex (flexor group — pulling inward). **Synergy 3** loads strongly on El.Flex and El.Ext together (independent **elbow modulation** for fine-tuning reach distance and diagonal targets). In impaired subjects, Synergy 3 is **lost** (merged into the first two), so only 2 PCs are needed for >90% VAF. This parallels the loss of independent motor modules seen in Clark et al.'s post-stroke walking data.

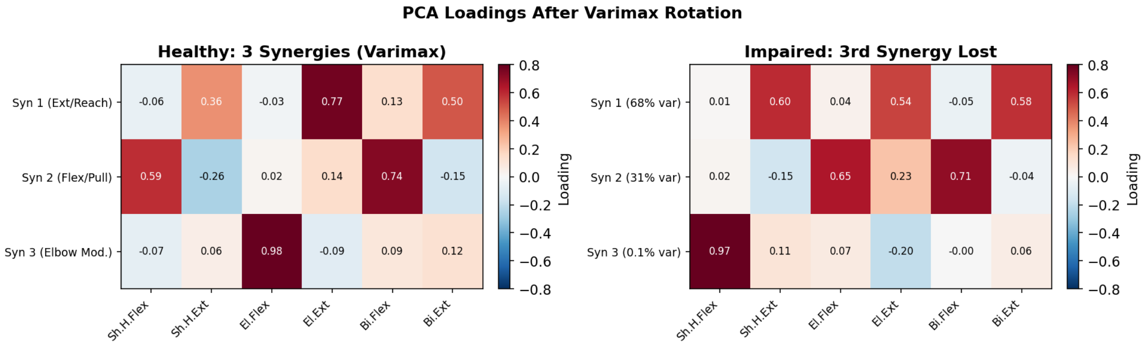

Figure 7. PCA loadings after Varimax rotation for healthy and impaired groups. Red = strong positive loading (muscle recruited by that synergy); blue = inhibited. Healthy: 3 distinct synergies each with clear muscle groupings — Syn 1 (Ext/Reach), Syn 2 (Flex/Pull), Syn 3 (Elbow Modulation, with El.Flex loading 0.98). Impaired: Synergies 1 and 2 absorb most variance; Synergy 3 explains only 0.1% (effectively lost), confirming that only 2 independent modules remain.

What would we see without Varimax rotation?

Without Varimax, unrotated PCA orders components strictly by variance explained. PC1 captures the *single direction of maximum spread* — typically a mixture of the extensor/flexor contrast and overall activation level. PC2 captures the next orthogonal direction, which is again a mathematical optimum rather than a physiologically clean grouping. The result is that unrotated loadings often show **mixed patterns**: a single PC may load on muscles from multiple functional groups simultaneously, making it hard to say 'this PC corresponds to the extensor synergy.' Varimax rotation redistributes variance among the retained PCs so that each loading vector has a few large weights and many near-zero weights — a **simple structure** — which aligns with the physiological intuition that synergies are discrete muscle groupings. The total variance explained by the retained set of PCs is unchanged; only its partition across components changes.

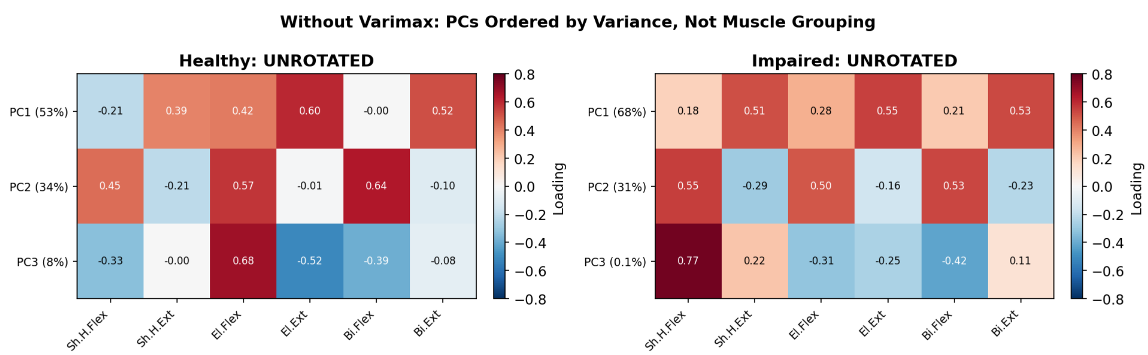

Figure 7b. Unrotated PCA loadings for comparison. Without Varimax, PC1 mixes multiple muscle groups because it simply maximizes variance. After rotation (Figure 7), each component aligns with a recognizable muscle grouping. Compare the two figures: the rotated synergies are easier to name and interpret physiologically.

In the **impaired** group, only **2 PCs are needed** for >90% variance (vs. 3 in healthy). The third synergy (independent elbow control) is **lost** — its variance has been absorbed into the first two merged modules. The impaired synergies show **broader co-activation**: each module recruits muscles from both joints simultaneously, reflecting reduced independence of neural drive (analogous to Clark et al.'s merged stance modules post-stroke).

> **Clinical Link: After stroke, patients show synergy merging — muscles that were independently controlled become abnormally coupled. Healthy reaching requires 3 independent modules; impaired control collapses this to 2. PCA detects this as fewer PCs needed for the same variance threshold, paralleling Clark et al.'s walking findings (4 modules → 2–3).**

7. scikit-learn PCA API

Python's scikit-learn library provides a clean, consistent API for PCA. The key workflow involves three steps: standardize, fit, and inspect.

Step 1: Standardize the data

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X) # zero mean, unit variance
```

Step 2: Fit PCA and transform

```
from sklearn.decomposition import PCA
pca = PCA(n_components=3) # keep top 3 PCs
scores = pca.fit_transform(X_scaled) # (240, 3)
```

Step 3: Inspect results

```
pca.explained_variance_ratio_ # [0.704, 0.296, 0.000]
pca.components_ # (3, 6) loadings matrix
pca.inverse_transform(scores) # reconstruct from PCs
```

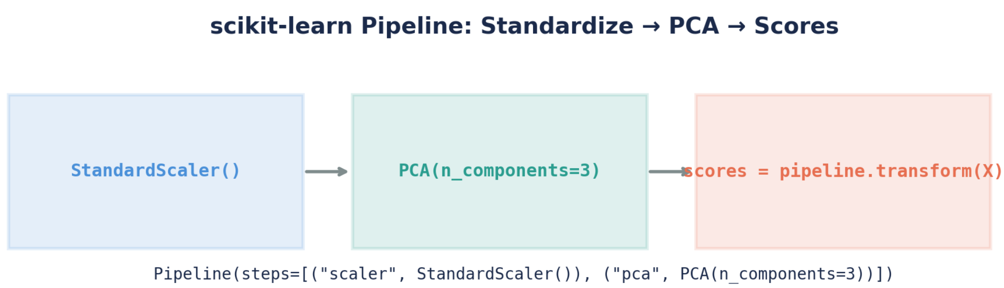

Figure 8. scikit-learn PCA workflow: fit_transform to reduce dimensionality, key attributes for interpretation, and the pipeline preview connecting to Week 5 classification.

An important preprocessing step is **standardization**. Without it, muscles with larger raw amplitude would dominate the PCA simply due to scale, not because they carry more task-relevant information. StandardScaler ensures each muscle contributes equally by centering to zero mean and scaling to unit variance.

> **API Pattern: scikit-learn follows a consistent fit/transform/predict API across all estimators. The PCA workflow you learn here transfers directly to classification (Week 5) and clustering (future weeks).**

8. Healthy vs. Impaired: Clinical Comparison

Projecting both groups into the **same PCA space** (fitted on healthy data) reveals how impairment alters the synergy structure. We use the healthy PCA to define the coordinate system, then project impaired data into those coordinates — this shows how impaired patterns deviate from the healthy norm.

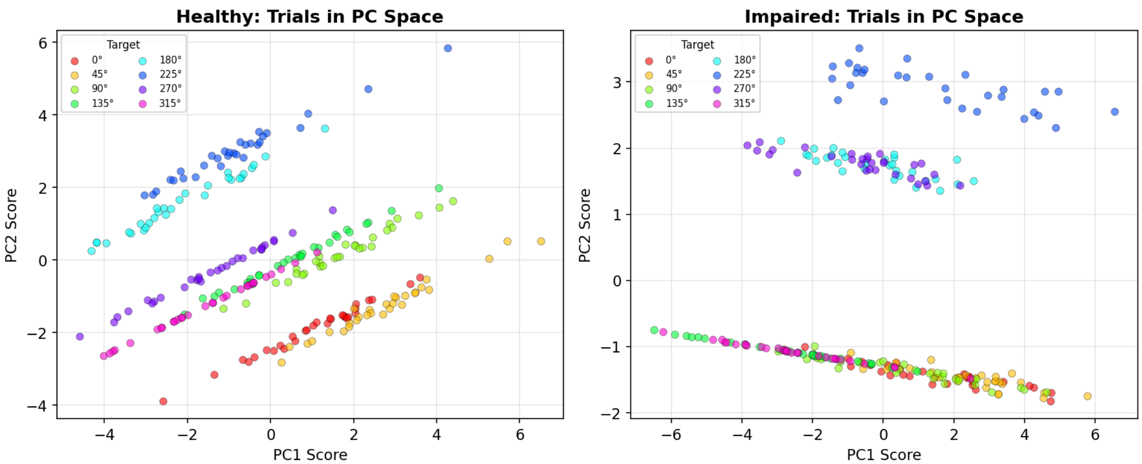

Figure 9. Each trial's PCA scores (PC1 vs PC2), colored by the known target direction (colors overlaid for visualization — PCA itself used no labels). Left: Healthy subjects show a broad spread with directions arranged progressively around the PC space, reflecting 3 independent synergies creating diverse muscle patterns. Right: Impaired subjects show trials compressed along a single diagonal axis — with only 2 synergies, directional control collapses toward a 1-D subspace. In Week 5, we will ask: can a classifier learn to decode direction from these patterns?

Two key observations (remember, PCA knew nothing about target direction — this structure emerged purely from variance in the muscle data):

- **Healthy spread:** Trials fan out across 2D PC space, with different directions occupying distinct regions — the 3 synergies create a rich, 2-dimensional representation of direction
- **Impaired compression:** Trials collapse toward a single diagonal, because 2 synergies can only generate a 1-D manifold of muscle patterns. Directions that were separable in 2D now overlap along this line

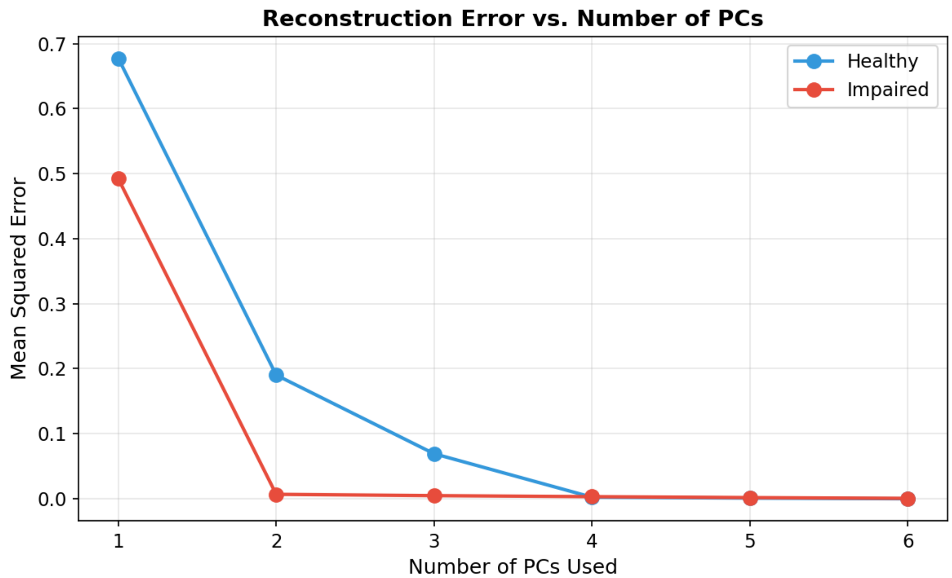

Figure 10. Reconstruction error vs. number of PCs. Healthy subjects need 3 PCs to approach near-zero error, while impaired subjects achieve similar accuracy with just 2 PCs. This difference in reconstruction curves directly reflects the loss of the third motor module.

Feature	Healthy	Impaired
Synergy structure	3 independent modules (3 PCs)	2 merged modules (2 PCs)
PC1 loadings	Extensor-dominated direction pattern	Alternating flex/ext (similar to healthy PC1)
Directional selectivity	Well-separated clusters	Compressed, overlapping
PC1 variance	~57% (3 PCs needed for >90%)	~55% (2 PCs sufficient for >90%)
Clinical implication	Independent synergy control	Loss of modularity

Week 4 Takeaway: PCA is a powerful unsupervised tool for discovering structure in high-dimensional neural/muscle data. The synergies it reveals are both computationally useful (dimensionality reduction for ML) and neuroscientifically meaningful (they reflect how the CNS organizes motor control).

Motor Modules Beyond Reaching: Walking After Stroke

The synergy framework extends well beyond reaching. A landmark study by Clark et al. (2010, *J Neurophysiol*) applied the same logic to **walking**, recording EMG from 8 unilateral leg muscles in 20 healthy controls and 55 post-stroke survivors. Rather than PCA, they used **nonnegative matrix factorization (NNMF)** — a related decomposition method that constrains weights to be nonnegative, producing directly interpretable synergy patterns.

Key findings:

- **Healthy controls: 4 motor modules** were sufficient to reconstruct the 8-muscle EMG during walking. Each module corresponded to a distinct gait-phase function: (C1) weight acceptance in early stance, (C2) ankle push-off for propulsion, (C3) leg swing and ground clearance, and (C4) leg deceleration in late swing.
- **Post-stroke (paretic leg): 2–3 merged modules.** Modules that were independent in healthy controls became fused — for example, C1 and C2 merged into a single stance module, and C3 and C4 merged into a single swing module. This merging reflects loss of independent neural drive to different muscle groups.
- **Fewer modules = worse function:** The number of modules correlated with preferred walking speed, step length asymmetry, and propulsive force asymmetry.

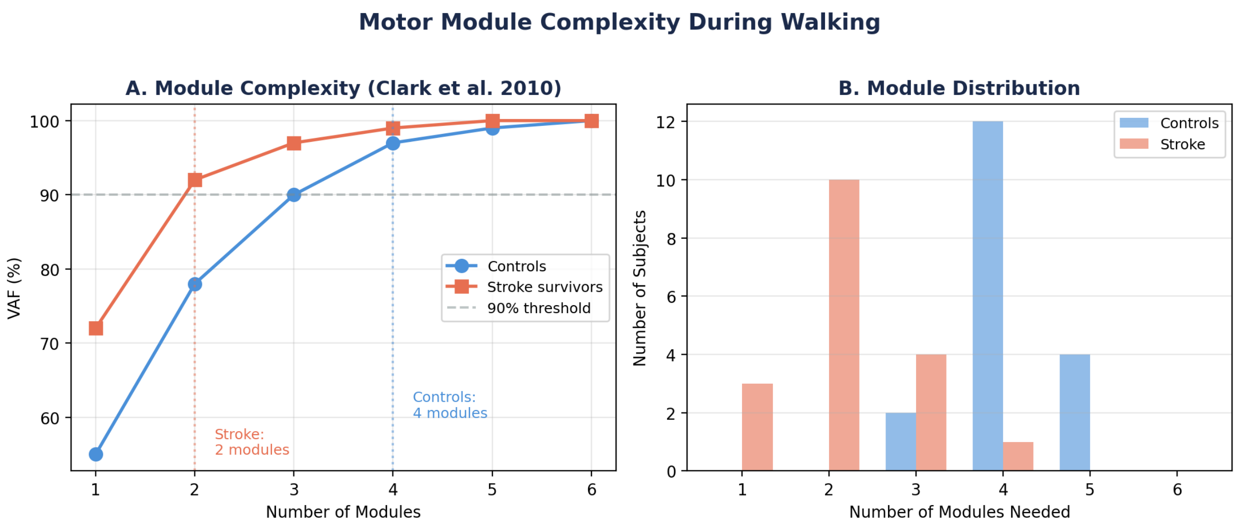

Figure 11. Motor module complexity during walking (based on Clark et al. 2010). A: Total variability accounted for (VAF) vs. number of motor modules. Paretic legs (red) reach 90% VAF with only 2 modules, while control legs (blue) need 4. The higher VAF for fewer modules in the paretic leg indicates less complex muscle coordination. B: Distribution of module number across participants. 45% of paretic legs required only 2 modules vs. 55% of controls requiring 4. 8 unilateral leg muscles: TA, SOL, MG, VM, RF, MH, LH, GM. Motor modules extracted via NNMF.

This walking example illustrates two important points. First, the motor module concept generalizes across different tasks (reaching, walking, posture) and muscle sets — wherever the CNS faces motor redundancy, low-dimensional structure emerges. Second, neurological injury consistently reduces the number of independent modules, and this reduction has direct functional consequences. Whether we use PCA or NNMF, the core insight is the same: dimensionality reduction reveals the control architecture of the nervous system.

Note: NNMF constrains synergy weights to be nonnegative (unlike PCA, which allows negative loadings). This often produces more physiologically interpretable results for EMG data, since muscle activations are inherently nonnegative. We use PCA in this course because it is more general and connects to the broader ML dimensionality reduction toolkit.

9. Looking Ahead: From Structure to Prediction

Throughout this week, we used PCA as an **unsupervised** tool: it found structure in the muscle data without knowing anything about target directions or group labels. The synergy structure, the scree plots, the loadings — all emerged purely from variance in the data.

In the lab, you projected **all 480 trials** (both healthy and impaired) into a shared PCA space and colored them by target direction:

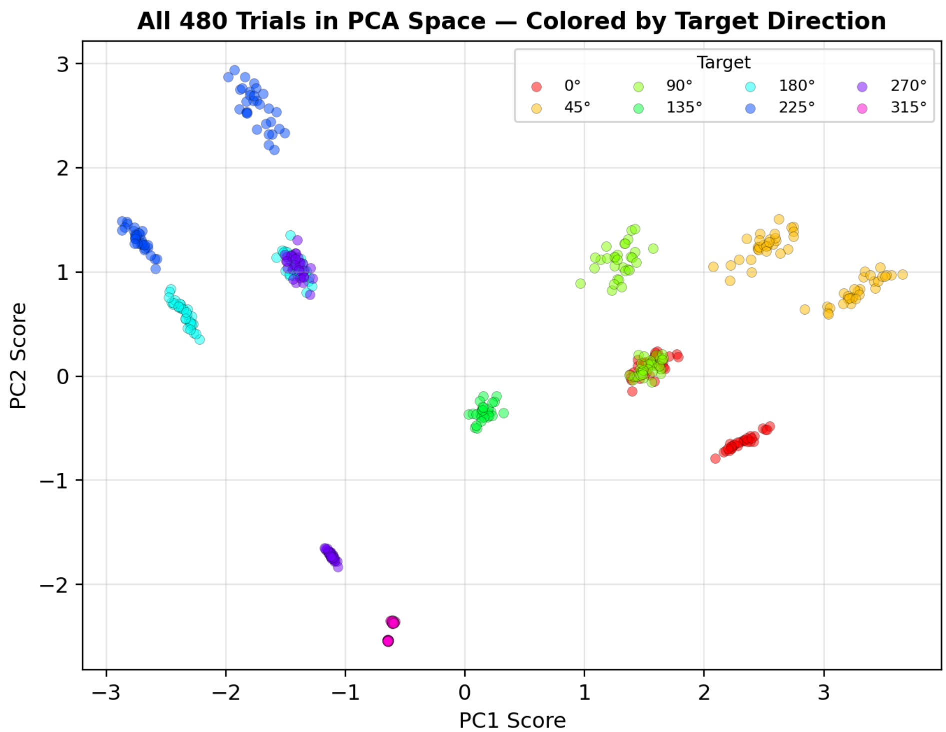

Figure 12. All 480 trials projected into a shared PCA space, colored by target direction. For several directions, you can see what appear to be two sub-groups within each cluster. PCA found this directional structure without any labels — but what explains the sub-groups?

Notice that for each target direction, there appear to be two sub-groups of trials. In Week 5, we will re-color these same points by group label (healthy vs impaired) and discover that the sub-groups correspond to the two populations. The impaired trials are systematically shifted relative to the healthy trials for the same direction — a signature of altered synergy structure. Can a classifier learn to exploit this offset? That is the question we take up next.

- **Direction decoding:** Can a supervised algorithm learn to predict which of the 8 directions a trial was aimed at, using these PCA features as inputs?
- **Clinical diagnosis:** Can a classifier separate healthy from impaired trials, given that both groups produce similar directional patterns?

These are **classification** problems — the shift from unsupervised learning (asking 'what structure exists?') to **supervised** learning (asking 'what predicts the label?'). That is what we take up in Week 5.

Week 5 Preview: Classification Fundamentals

Using PCA scores as input features, we will build logistic regression classifiers for two tasks: (1) decoding which of 8 directions a trial targeted, and (2) diagnosing whether a subject is healthy or impaired. We will learn about the sigmoid function, regularization (L1 vs L2), cross-validation pitfalls (why k-fold on trials overestimates accuracy), and clinical evaluation metrics (ROC curves, sensitivity vs specificity). The PCA scores from this week become next week's classifier inputs.